Getting Started with OpenLDAP

Part 1

Setting Up A Directory Server

Let us begin this series on LDAP by setting up openLDAP on our machines and unravelling its mysteries as the series progresses.

OpenLDAP is the open source implementation of the LDAP protocol to access a directory. Simply put, a directory is a repository of data, much like a database (but with significant differences) that is used to store huge amounts of data. Directories are optimised for fast search and retrieval. Directories are also fairly static—in the sense that the data in a directory is not updated very frequently.

LDAP or 'Lightweight Directory Access Protocol' is a protocol used to access directories. OpenLDAP is an open source directory software suite conforming to the LDAP protocol (version 3) and it supports all major platforms. The current version of OpenLDAP is 2.4. There are significant changes in this version compared to v 2.3. In this article though, we will cover features that apply to the previous versions as well.

The OpenLDAP software suite consists of a directory server (*openldap-server*) and a client (*openldap-clients*) to access the directory. To be able to query other LDAP servers, only the *openldap-client* is required. The *openldap-client* package provides popular client tools such as *ldapsearch*, *ldapadd*, etc. The *openldap-server* software is required if you want to build a directory of your own.

On a Red Hat Enterprise Linux/Fedora system, the OpenLDAP RPMS are available as *openldap-server* and *openldap-clients*. A simple check will reveal the RPMS installed in your system. To query the RPM database for the installed OpenLDAP RPMS, run the following command:

```
[vbg@vbg-work scratch]$ rpm -qa | grep openldap
```

In my system, the output was as follows: